

Qatar-1b

R-2728

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110823 · created 2026-09-02T00:00:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455796.73492 +/- 0.0004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.65 +/- 2.13 [%]
Orbital Inclination (inc)	87.05 +/- 2.81
Airmass coefficient 1 (a1)	0.022 +/- 0.014
Airmass coefficient 2 (a2)	2.99 +/- 1.61
Scatter in the residuals of the lightcurve fit is	61.36 %
Best Comparison Star	#2 - [478, 137]
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.056 vs 0.1463 (deviation 0.66 σ)
Duration fit vs published	0.08 d vs 0.0692 d (deviation 0.81 σ)
Mid-time O-C	-1.0 min
Depth SNR	2.8
Residual scatter	61.36 %

Light curve

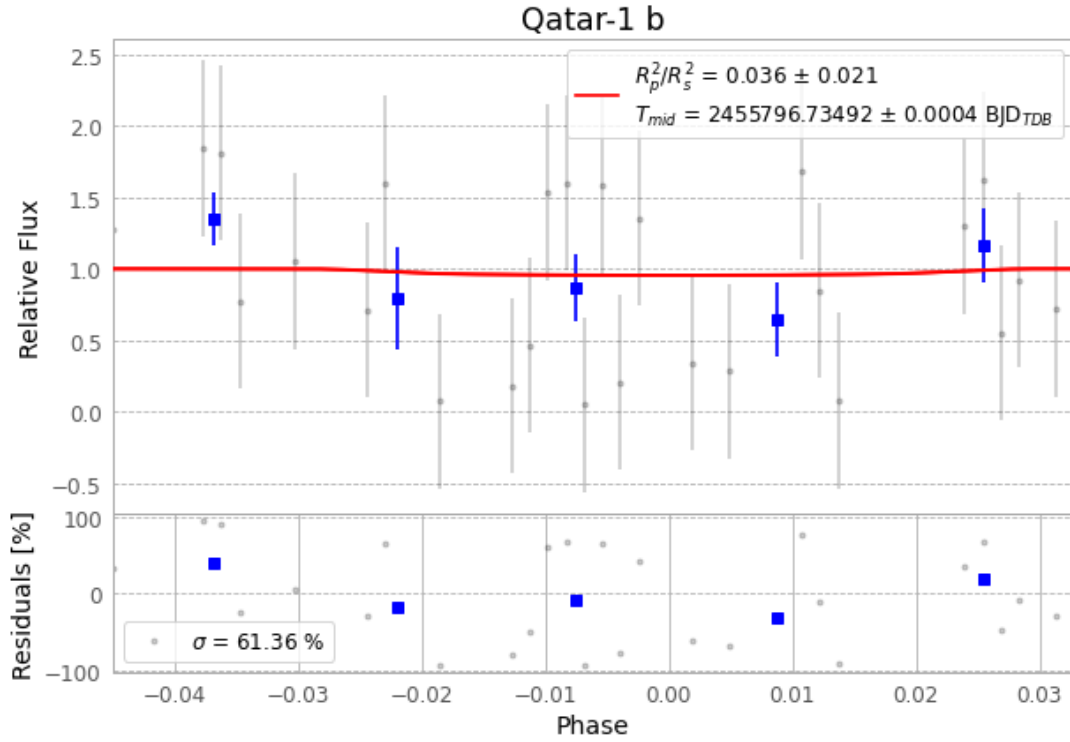


Figure 1 — FinalLightCurve_Qatar-1 b_2011-08-22.png (SHA-256 0247277e33a48372...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [243, 383], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 55713acc98234ba5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), Qatar-1110823040043.FITS → Qatar-1110823073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110823.log (21752be672cf...), FinalLightCurve_Qatar-1 b_2011-08-22.png (0247277e33a4...), FinalLightCurve_Qatar-1 b_2011-08-22.csv (5888231aca54...)

Compute resources

Wall time 167.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:00Z.